Photosynthesis: How Plants Make Food

What photosynthesis is

Photosynthesis is the process by which green plants, algae and some bacteria use light energy to make glucose from carbon dioxide and water. Oxygen is released as a by-product. The overall word equation is: carbon dioxide + water, in the presence of light and chlorophyll, gives glucose + oxygen. The balanced symbol equation is 6CO2 + 6H2O -> C6H12O6 + 6O2. Photosynthesis is an endothermic reaction: it takes in energy from the surroundings, in the form of light.

Where it happens

Photosynthesis takes place inside chloroplasts, small organelles found mainly in the cells of the leaf, especially the palisade mesophyll layer just below the upper surface. Chloroplasts contain the green pigment chlorophyll, which absorbs mostly red and blue light and reflects green light, which is why leaves look green. Carbon dioxide enters the leaf through tiny pores called stomata, usually on the underside of the leaf. Guard cells open and close the stomata. Water is absorbed by the roots and carried up to the leaves in xylem vessels.

The two stages

Photosynthesis happens in two linked stages. In the light-dependent reactions, which take place on the thylakoid membranes, light energy splits water molecules (photolysis). This releases oxygen and produces the energy carriers ATP and NADPH. In the light-independent reactions, also called the Calvin cycle, which take place in the stroma, ATP and NADPH are used to fix carbon dioxide into glucose. The enzyme RuBisCO catalyses the first step of carbon fixation. The light-independent stage does not need light directly, but it stops soon after dark because it depends on the products of the light-dependent stage.

Limiting factors

The rate of photosynthesis is controlled by limiting factors: light intensity, carbon dioxide concentration and temperature. A limiting factor is whichever factor is in shortest supply; increasing it raises the rate until another factor becomes limiting. Temperature affects the enzymes involved: the rate rises with temperature up to an optimum (around 25 to 35 degrees Celsius for many plants) and then falls quickly as the enzymes denature. Farmers growing crops in greenhouses can raise all three factors, for example with paraffin heaters that give off both heat and carbon dioxide, to increase yield.

What the plant does with glucose

Plants use the glucose made in photosynthesis in several ways. Some is used immediately in respiration to release energy. Some is converted into insoluble starch for storage, which is why a leaf kept in light tests positive for starch with iodine solution (turning blue-black). Glucose is also used to make cellulose for cell walls, converted into lipids for storage in seeds, and combined with nitrate ions absorbed from the soil to make amino acids, which build proteins.

Photosynthesis and respiration

Photosynthesis and aerobic respiration are roughly opposite processes. Respiration uses glucose and oxygen and releases carbon dioxide, water and energy; it happens all the time, day and night, in every living cell. At low light, such as at dawn and dusk, a plant can reach the compensation

point, where the rate of photosynthesis exactly equals the rate of respiration, so there is no net exchange of oxygen or carbon dioxide with the air.